Answer **all** *questions.*

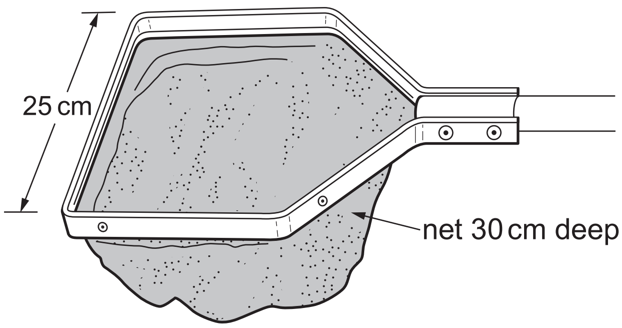
1. In Wales, 12 000 out of 24 000 km of rivers are estimated to be acidified (having a pH of less than 5.6).

Forests capture acidic pollutants from the atmosphere and release them into stream water.

A group of students investigated the effect of acidification on the biodiversity of freshwater invertebrates in streams in Mid Wales.

Kick sampling was used to compare the biodiversity of a moorland stream (pH = 6.5) and a forest stream (pH = 5.0).

The diagrams below give the dimensions of a “D net” and its use when sampling.



The kick sampling method used by the students is described below:

- Place the bottom edge of the net on the stream bed on the downstream side of the sampling point.
- Kick into the stones just upstream of the net, allowing the disturbed material to drift downstream and be caught in the net.
- Empty the contents of the net into a tray containing stream water.
- Identify and count the invertebrates.
- Return the invertebrates gently to the stream.

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(a) (i) Both streams were sampled in shallow, fast flowing regions. The samples were taken from areas of the same width and water depth, with similar stony stream beds.

Suggest **two** other variables that would need to be as similar as possible between the two streams. [2]

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(ii) When kick sampling, state **two** factors that need to be controlled to ensure standardisation of sampling. [2]

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(iii) To improve the accuracy of species identification it was suggested that the specimens collected could be preserved in alcohol and taken back to the laboratory for closer examination. Discuss why this was considered to be unethical. [2]

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2400U201 03

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(b) Five kick samples were obtained from each stream. The results are shown below.

Species of invertebrate (common names)	Total number of organisms of each species	
	Moorland stream (pH = 6.5)	Forest stream (pH = 5.0)
Caddisfly larva	8	1
Stonefly nymph	10	37
Wandering snail	3	0
Swimming beetle larva	2	8
Freshwater shrimp	39	0
Mayfly nymph	22	2
Total	84	48

(i) The students collected invertebrates at sites chosen at random. Explain the importance of the sites being chosen at random. [1]

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(ii) Suggest **two** reasons why the values obtained using this technique might be an underestimate of the actual numbers of species at the kick sample sites. [2]

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(iii) Use the following table and the formula to calculate Simpson's Diversity Index for the moorland stream. [3]

Species of invertebrate (common names)	n	$(n–1)$	$n(n–1)$
Caddisfly larva	8	7	56
Stonefly nymph	10	9	90
Wandering snail	3	2	6
Swimming beetle larva	2	1	2
Freshwater shrimp	39	38	1482
Mayfly nymph	22	21	462
	$N = 84$		$\Sigma n(n-1) =$ ....
	$N(N-1) =$		

N = total number of individuals of all species
n = number of individuals per species of each species
Σ = sum of

Simpson's Diversity Index $$D = 1 - \frac{\Sigma n(n-1)}{N(N-1)}$$

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(iv) The Diversity Index for the forest stream is 0.4. With reference to the results table on page 4, explain how you could deduce that the forest stream was less diverse than the moorland stream without needing to calculate the Diversity Indices. [2]

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(c) The students concluded that forests reduced species diversity in streams.

(i) Explain how the confidence in this conclusion could be improved. [2]

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(ii) Suggest how forest managers could use this information to increase diversity in forest streams. [1]

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17

2400U201 05